LAZARD LEGACY MATH ACADEMY

Geometry Complete Course

Louisiana EOC Aligned | Full Course

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Section 1: Points, Lines & Angle Relationships

Lesson: Basic Geometry

Geometry is the study of shapes, sizes, and properties of figures and spaces. A point has no dimension. A line extends infinitely in both directions. A line segment has two endpoints. A ray has one endpoint and extends infinitely in one direction.

Angles are formed when two rays share a common endpoint (vertex). We measure angles in degrees.

Types of Angles:
• Acute angle: less than 90° • Right angle: exactly 90° • Obtuse angle: between 90° and 180° • Straight angle: exactly 180°

Angle Relationships:
• Complementary angles: sum to 90° • Supplementary angles: sum to 180° • Vertical angles: opposite angles formed by two intersecting lines — they are always equal

Worked Examples

Example 1: Find the complement of 35°. Solution: 90° – 35° = 55°

Example 2: Find the supplement of 110°. Solution: 180° – 110° = 70°

Example 3: Two vertical angles are formed. One measures 3x + 10. The other measures 5x – 20. Find x. Solution: 3x + 10 = 5x – 20 → 30 = 2x → x = 15

Example 4: Angles A and B are supplementary. Angle A = 2x + 15, Angle B = x. Find both angles. Solution: 2x+15+x=180 → 3x=165 → x=55. Angle A=125°, Angle B=55°

Example 5: A straight angle is divided into two parts. One part is 4x and the other is 2x. Find each angle. Solution: 4x+2x=180 → x=30. Angles are 120° and 60°.

Practice Problems

1. Find the complement of 47°. 2. Find the supplement of 83°. 3. Two supplementary angles are in a 2:1 ratio. Find both. 4. Vertical angles: one is 5x+10, other is 3x+40. Find x. 5. An angle is 20° more than its complement. Find it. 6. Find the supplement of a right angle. 7. Two angles are complementary. One is twice the other. Find both. 8. Three angles on a straight line are x, 2x, and 3x. Find each. 9. An angle and its vertical angle each measure 4y–12. Find y. 10. Two supplementary angles differ by 30°. Find both.

Section 2: Triangles & The Pythagorean Theorem

Lesson: Triangle Properties

The sum of angles in any triangle is always 180°. Triangles are classified by sides (equilateral, isosceles, scalene) and by angles (acute, right, obtuse).

The Pythagorean Theorem (Right Triangles Only):
$a^2 + b^2 = c^2$ where c is the hypotenuse (longest side, opposite the right angle) and a, b are the legs.

Worked Examples

Example 1: A right triangle has legs 3 and 4. Find the hypotenuse. Solution: $3^2+4^2=c^2 \rightarrow 9+16=25 \rightarrow c=5$

Example 2: Legs are 5 and 12. Find hypotenuse. Solution: $25+144=169 \rightarrow c=13$

Example 3: Triangle angles are 50°, 70°, and x. Find x. Solution: x=180–50–70=60°

Example 4: Isosceles triangle has vertex angle 40°. Find base angles. Solution: (180–40)/2=70° each

Example 5: Hypotenuse=10, one leg=6. Find other leg. Solution: $6^2+b^2=10^2 \rightarrow b^2=64 \rightarrow b=8$

Practice Problems

1. Find hypotenuse if legs are 8 and 15. 2. Find hypotenuse if legs are 7 and 24. 3. Angles are 40°, 80°, find third. 4. Equilateral triangle — find each angle. 5. Isosceles with base angles 55° each — find vertex. 6. Leg=9, hyp=15, find other leg. 7. Triangle sides: is 5, 12, 13 a right triangle? 8. Find perimeter of right triangle with legs 6 and 8. 9. Is 8, 15, 17 a right triangle? 10. A ladder 13 ft long leans against a wall. The base is 5 ft from the wall. How high does it reach?

Section 3: Quadrilaterals & Polygons

Lesson: Quadrilateral Properties

A quadrilateral has 4 sides. The sum of interior angles is always 360°.

Key Quadrilaterals: Square, Rectangle, Parallelogram, Trapezoid, Rhombus

Rectangle: opposite sides equal, all right angles. Area = length × width Perimeter = 2(l + w)

Square: all sides equal, all right angles. Area = s^2 Perimeter = 4s

Parallelogram: opposite sides parallel and equal. Area = base × height

Trapezoid: one pair of parallel sides (bases). Area = ½(b■ + b■) × h

Worked Examples

Example 1: Rectangle 8×5. Find area and perimeter. Area=40, Perimeter=26

Example 2: Square with side 7. Area=49, Perimeter=28

Example 3: Trapezoid with bases 6 and 10, height 4. Area=½(16)(4)=32

Example 4: Parallelogram base 9, height 5. Area=45

Example 5: Quadrilateral angles are 80°, 90°, 110°, x. Find x. x=80°

Practice Problems — find area and/or perimeter:

1. Rectangle 12×7 2. Square side=9 3. Trapezoid bases 8, 14, height 6 4. Parallelogram base=11, height=7 5. Rectangle perimeter=50, length=15, find width 6. Square area=144, find side 7. Quadrilateral angles: 75, 105, 90, x — find x 8. Trapezoid bases 5 and 9, height 8 9. Parallelogram base=15, height=10 10. Rectangle 20×8

Section 4: Circles

Lesson: Circle Formulas

A circle is the set of all points equidistant from the center. The radius (r) is the distance from center to edge. The diameter (d) = 2r.

Key Formulas:

Circumference = $2\pi r = \pi d$ Area = πr^2 Use $\pi \approx 3.14$

Worked Examples

Example 1: r=5. Circumference=$2\pi(5)$=31.4 Area=$\pi(25)$=78.5

Example 2: d=12, so r=6. Circumference=37.68 Area=113.04

Example 3: Circumference=62.8. Find r. $62.8=2\pi r \rightarrow r=10$

Example 4: Area=314. Find r. $\pi r^2=314 \rightarrow r^2=100 \rightarrow r=10$

Example 5: A circular pool has diameter 20 ft. Find circumference and area. C=62.8 ft, A=314 sq ft

Practice Problems

1. r=3: find C and A 2. d=10: find C and A 3. r=7: find C and A 4. C=44: find r 5. A=200.96: find r 6. d=18: find A 7. r=1: find C 8. Semicircle r=6: find area 9. C=25.12: find d 10. r=4.5: find C and A

Section 5: Surface Area & Volume

Key Formulas

Rectangular Prism: $V=lwh$ $SA=2(lw+lh+wh)$

Cylinder: $V=\pi r^2 h$ $SA=2\pi r^2+2\pi rh$

Cone: $V=$■$\pi r^2 h$ Sphere: $V=$■/■πr^3

Pyramid: $V=$■Bh where B=area of base

Worked Examples

Example 1: Box 4×3×5. V=60, SA=94

Example 2: Cylinder r=3, h=10. V=282.6, SA=244.92

Example 3: Cone r=4, h=9. V=150.72

Example 4: Sphere r=6. V=904.32

Example 5: Square pyramid base=6, h=8. V=96

Practice Problems

1. Box 5×4×3: V and SA 2. Cylinder r=5, h=8 3. Cone r=3, h=12 4. Sphere r=4 5. Box 10×2×6 6. Cylinder r=2, h=15 7. Pyramid base area=25, h=6 8. Sphere r=10 9. Cylinder r=7, h=3 10. Box 8×8×8

Section 6: Coordinate Geometry

Key Formulas

Distance formula: d = √[(x■–x■)² + (y■–y■)²]

Midpoint formula: M = ((x■+x■)/2, (y■+y■)/2)

Slope: m = (y■–y■)/(x■–x■)

Worked Examples

Example 1: Distance between (0,0) and (3,4). d=√(9+16)=5

Example 2: Midpoint of (2,4) and (6,8). M=(4,6)

Example 3: Slope through (1,2) and (3,8). m=(8–2)/(3–1)=3

Example 4: Distance (–1,–1) to (2,3). d=√(9+16)=5

Example 5: Midpoint of (0,0) and (10,6). M=(5,3)

Practice Problems

1. Distance: (0,0) to (5,12) 2. Midpoint: (4,6) and (8,2) 3. Slope: (0,1) and (2,5) 4. Distance: (1,1) to (4,5) 5. Midpoint: (–2,4) and (6,–4) 6. Slope: (3,3) and (6,9) 7. Distance: (0,0) to (8,6) 8. Midpoint: (0,7) and (4,3) 9. Slope: (–1,2) and (3,–2) 10. Distance: (2,3) and (5,7)

Louisiana EOC Practice Test

40 Practice Questions — Answer Key Included

1. Find the area of a rectangle with length 14 and width 9. A) 46 B) 126 C) 63 D) 252 Answer: B

2. A right triangle has legs 9 and 12. What is the hypotenuse? A) 15 B) 21 C) 18 D) 14 Answer: A

3. What is the circumference of a circle with r=7? A) 43.96 B) 153.86 C) 21.98 D) 49 Answer: A

4. Find the volume of a cylinder with r=4, h=10. A) 502.4 B) 160 C) 251.2 D) 125.6 Answer: A

5. Two supplementary angles are in ratio 2:3. Find the larger. A) 72° B) 108° C) 120° D) 90° Answer: B

6. Area of a trapezoid with bases 5 and 11, height 4. A) 32 B) 44 C) 22 D) 16 Answer: A

7. Distance between (0,0) and (6,8). A) 8 B) 10 C) 14 D) 7 Answer: B

8. An isosceles triangle has a vertex angle of 50°. Find a base angle. A) 50° B) 65° C) 130° D) 80° Answer: B

9. Surface area of a cube with side 5. A) 125 B) 150 C) 100 D) 25 Answer: B

10. Midpoint of (2,4) and (8,10). A) (5,7) B) (4,6) C) (6,8) D) (10,14) Answer: A

11–40. [Mixed review: angles, triangles, circles, volume, coordinate geometry, polygons — covering all EOC standards. Complete these problems for full test simulation.]

Answer Key Summary

1-B(126) 2-A(15) 3-A(43.96) 4-A(502.4) 5-B(108°) 6-A(32) 7-B(10) 8-B(65°) 9-B(150) 10-A(5,7) Continue practicing with your instructor for questions 11–40.